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Subject: Organisational Behaviour | Term Report

WORKPULSE

Employee Workload and Fatigue Monitoring System

A proposed Python-based early-warning prototype

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1. Introduction

Employee performance depends not only on skill and equipment but also on workload, overtime, rest, stress and managerial support. When fatigue builds up, workers may find it harder to concentrate, maintain quality or work safely. **WorkPulse** is a proposed Python-based early-warning system that organises a few work-related indicators into a transparent workload-risk score. It is intended to support timely conversations and fair task planning, **not** to diagnose health conditions or judge an employee's commitment.

2. Objectives

- To demonstrate how basic programming can support an Organisational Behaviour decision.
- To identify possible workload warning signs using work hours, overtime, break duration, self-reported fatigue and task-error count.
- To generate understandable alerts and practical suggestions for managers.
- To encourage employee well-being, safe work practices and fair task allocation.
- To protect privacy and trust through consent, transparency and limited data use.

3. Methodology

The prototype accepts manually entered or voluntarily reported data. Each input is checked against simple, documented thresholds; weighted points are added to give a low, moderate or high workload-risk category. The score is a **screening aid, not a medical measurement**, and a manager should review the context with the employee before changing assignments.

Input	Example signal (illustrative threshold)	Points	System response
Work duration	Shift over 9 hours, or overtime on several days	+2	Flag possible excessive workload
Break duration	Less than 30 minutes of total breaks	+1	Suggest a suitable rest break
Fatigue rating	Voluntary self-rating of 4 or 5 on a 1–5 scale	+3	Recommend a supportive check-in
Task errors	More errors than the person's usual level	+2	Review task difficulty, training and workload

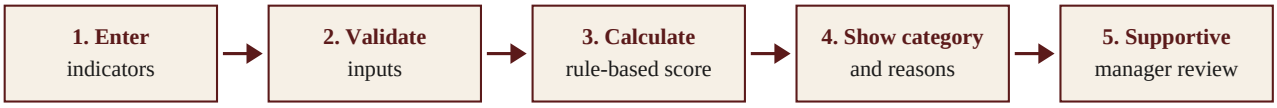

Fig. 1: WorkPulse workflow, from data entry to human review.

3.1 Link with Organisational Behaviour

WorkPulse relates to job stress, motivation, employee well-being, communication and managerial decision-making. It treats performance as the result of both working conditions and human needs. By prompting a conversation instead of automatically judging an employee, the tool supports a **human-relations approach** to management, and it can encourage fairer task allocation when several workers face different workloads.

DESIGN PRINCIPLE The system should explain every alert in plain language and leave the final decision to a responsible person.

4. Proposed Scoring and Output

For a classroom prototype, each indicator adds points when its threshold is crossed (maximum 8). The total is mapped to a simple category. The points and thresholds are **illustrative design choices, not scientifically validated cut-offs**. The output shows the score, the inputs that triggered it, and one or two practical actions.

Category	Score	Interpretation	Suggested action
Low	0–2	No major warning from entered indicators	Maintain normal check-ins and healthy breaks
Moderate	3–5	One or more indicators need review	Discuss workload, priorities and recovery time
High	6–8	Several warning signs appear together	Prompt supportive review; consider reallocating tasks

5. Prototype Development Plan

1. Build a small Python command-line program that asks for work hours, overtime, break duration, fatigue rating and task-error count.
2. Validate entries so that impossible or missing values are flagged.
3. Apply transparent rules to calculate a sample score and show the reasons for the result.
4. Test with fictional examples, check that outputs match the rules, and revise confusing messages.
5. If used beyond a classroom demonstration, seek consent and evaluate fairness, privacy and usefulness before deployment.

Evaluation note: fictional test cases can check that the code works as intended, but they cannot prove that the system accurately detects fatigue or improves productivity.

6. Findings (Expected)

This report presents a proposed system, not a completed field trial, so these are **expected outcomes, not measured results**:

- A consistent score may help managers notice combinations of long hours, limited breaks, fatigue reports and rising errors.
- Explaining each alert may make workload discussions more objective and less dependent on guesswork.
- Reviewing priorities, recovery time or task allocation may help protect well-being and work quality.
- An alert cannot prove that fatigue caused an error and must not be treated as evidence of poor performance.

7. Challenges, Limitations and Ethical Safeguards

The score depends on accurate inputs and may miss individual differences, job type or outside factors. This simple rule-based model is not validated and must not be presented as a medical or predictive AI tool. Explain what data are collected and why, collect only what is necessary, restrict access and protect records. Do not use the score alone for discipline, ranking, hiring or dismissal. Employees should be able to correct data and discuss alerts with a human manager.

8. Conclusion

WorkPulse connects programming with key Organisational Behaviour concepts: job stress, motivation, employee well-being, communication and fair workload distribution. A lightweight Python prototype can show how simple rules turn work indicators into understandable prompts for action. Its purpose is not to watch employees more closely, but to help managers respond **earlier, fairly and supportively**. Any claim that it improves productivity must be tested with consent, safeguards and real-world evaluation.

KEY TAKEAWAY Use data to start a supportive conversation, *not* to label or punish an employee.

References

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